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Original Article

Multidimensional characterization of craniofacial skeletal phenotype of obstructive sleep apnea in adults

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Abstract

Objectives: We aimed to characterize a craniofacial skeletal phenotype (CSP) of adult obstructive sleep apnea (OSA) patients from a multidimensional perspective, exploring the impact of transverse skeletal discrepancy (TSD) on multivariable polysomnographic profiles.

Materials and methods: This retrospective, cross-sectional study included 102 adult OSA patients. Sagittal, vertical, and transverse skeletal patterns were categorized on the cone beam computed tomography images. The CSP of OSA patients, characterized by a Class II hyperdivergent pattern, was divided into CSP_{2D} and CSP_{3D} subgroups according to the presence of TSD, and compared with the non-CSP of OSA patients. Both nasal and pharyngeal airway variables were involved for assessment, and 12 polysomnographic variables with a sleepiness symptom variable were used for phenotype-based inter-group comparisons.

Results: The CSP patients revealed greater disease severity than the non-CSP patients (indicated by eight polysomnographic variables), despite being younger ($P < .05$) and less obese ($P < .01$). The CSP_{3D} patients with TSD exhibited more severe OSA than the age- and BMI-matched CSP_{2D} patients without TSD, as indicated by nine polysomnographic variables, in relation to smaller nasal airway volume, smaller pharyngeal minimum cross-sectional area, and longer pharyngeal airway length (all $P < .05$). The probability of multiperspective characteristics among three phenotypes was significantly contrasted in 19 variables.

Conclusions: From a multidimensional perspective, the CSP patients showed greater OSA severity with more vulnerable nasal and pharyngeal airways than non-CSP patients, despite being younger and less obese. Specifically, the CSP_{3D} patients revealed far more severe OSA than the CSP_{2D} patients, indicating the significance of TSD on the polysomnographic profiles.

Keywords: craniofacial phenotype; obstructive sleep apnea; polysomnography; skeletal pattern; transverse discrepancy; upper airway; cone-beam computed tomography

Introduction

Successful management of obstructive sleep apnea (OSA) still remains a challenge, despite advances in diagnostic technologies and therapeutic modalities, due to its heterogeneity and complexity in pathophysiologic pathways, clinical symptoms, comorbidities, and therapeutic responses [1]. With updated insights into OSA phenotyping [2], the importance of differential diagnosis on the main phenotypic cause of OSA underlines the need of tailoring personalized treatment for each patient [3]. OSA often originates from an anatomically collapsible upper airway (UA), though it may be combined with nonanatomical pathologic perpetuation by poor muscle responsiveness, low arousal threshold, or oversensitive ventilator control [3]. Considering that the anatomically collapsible UA is primarily influenced by the relationship between craniofacial skeletal structure and UA soft tissues [4], an in-depth investigation of the craniofacial anatomical features contributing to OSA is crucial for a more integrative interpretation of multidimensional disease profiles.

Recently, cone beam computed tomography (CBCT) has been increasingly utilized for three-dimensional (3D) measurement of the UA. This modality has demonstrated high accuracy in evaluating not only pharyngeal volume but also the minimal cross-sectional area of these regions making it a valuable tool in both clinical and research settings [5–7]. The latest software programs for UA evaluation have also proven to be highly reliable, particularly in assessing oropharyngeal volume and minimal cross-sectional area [8]. Furthermore, advancements in AI algorithms have enabled the automatic segmentation of anatomically collapsible UAs, thereby enhancing the utility of these assessments in OSA patients [9].

Morphological investigations of craniofacial predisposing factors for OSA have evolved from traditional two-dimensional (2D) cephalometric studies to 3D CBCT studies. Most commonly, 2D craniofacial factors vulnerable to OSA have been enumerated through a variable-to-variable analysis, such as short cranial base length, retrusive small mandible, depressed or constricted maxilla, hyperdivergent vertical pattern, and

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lower position of the hyoid, along with pharyngeal airway narrowing and lengthening [4, 10].

Even CBCT studies still mostly use multi-planar analysis to assess the impact of skeletal malocclusion on UA measurements in one-dimensional or 2D aspects, often overlooking information from other planes [6, 8, 11–14]. Few studies evaluated the impact of transverse skeletal discrepancy (TSD) on the disease severity in conjunction with sagittal and vertical skeletal patterns, going beyond image analysis to characterize the examined patient. In this context, the characterization of OSA patients who have skeletal vulnerabilities in 3D was anticipated to identify a craniofacial skeletal phenotype (CSP) of OSA.

To categorize the OSA phenotypes, medical researchers have conducted a cluster analysis using objective OSA severity and/or subjective symptoms as principal components, without consideration of craniofacial factors [3, 15–19]. We had previously performed a cluster study incorporating sagittal and vertical skeletal variables into the OSA severity and symptom variables. This approach enabled the differentiation of a CSP with a skeletal Class II hyperdivergent pattern from a nonanatomical or complex phenotype [15, 16]. Another recent cluster study [17] attempted to characterize three OSA phenotypes, including CBCT skeletal and UA variables in multiple polysomnographic variables. However, this study could neither identify a CSP nor discriminate the 3D craniofacial features among clusters. Despite the clinical significance of cluster studies, the generated clusters vary depending on the sample size or selected analytical components even within the same population. Moreover, the outcome clusters may not encompass every OSA patient. Therefore, integrative research is needed to characterize the CSP of OSA patients from a multidimensional perspective to assist orthodontists in recognizing and effectively managing CSP in their clinical practice.

When identifying a CSP, the concurrent assessment of other anatomical risk factors such as submental fat deposition due to obesity and UA soft tissue abnormalities is essential [4]. Additionally, nonanatomical phenotypic factors such as upper airway muscle responsiveness, respiratory arousal threshold, and high loop gain causes may coexist [2]. Moreover, any risk factors associated with the phenotypic causes of OSA such as ethnicity, genetics, age, and gender should be considered during outcome interpretation [20–23]. This raised a clinical question regarding the existence of any significant skeletal predispositions to OSA severity that are independent of the patient's demographic baselines. Such insights can provide orthodontists with valuable clinical evidence to facilitate individual risk assessments and guide therapeutic target decisions for the CSP of OSA patients.

We aimed to characterize the CSP of adult OSA patients from a multidimensional perspective based on an integrative investigation of sagittal, vertical, and transverse skeletal discrepancies, region-specific whole UA features, and multivariable polysomnographic presentations considering demographic variances. The null hypothesis was that the CSP of OSA patients had no significant differences in disease profiles from the non-CSP of OSA patients, regardless of the accompanied TSD.

Materials and methods

Design and participants

In this retrospective cross-sectional study, the demographic and clinical data of 160 Korean adult OSA patients (aged 19–72 years old, apnea–hypopnea index (AHI) ≥ 5 events/h) who visited the department of Orthodontics in Kyung Hee University Dental hospital between January 2019 and December 2022 were reviewed from an electronic medical record. Patients who underwent full-night laboratory polysomnography and were diagnosed with OSA by a physician, as well as those who required detailed anatomical assessment for three-dimensional (3D) evaluation of severe skeletal discrepancies, multidisciplinary treatment planning, and orthognathic surgery planning, had craniofacial CBCT scans and completed the Epworth Sleepiness Scale (ESS) included in the study. Fifty-two patients who had central apnea, serious cardiopulmonary comorbidities, syndromic craniofacial deformity, a history of UA or skeletal surgery, a history of orthodontic or craniofacial orthopedic treatment, or poor-quality CBCT images taken in a non-standardized position were excluded. Six patients with skeletal Class III deformity were also excluded due to their notably low frequency (3.7%) for statistical reasons.

The sample size was calculated for inter-phenotypic comparison using G* Power Software (version 3.1.9.7, Düsseldorf, Germany), referring to the previous relevant studies [11, 24]. An effect size of 0.4, a power of 80%, and an α value of 0.05 were applied. The minimum required sample size was 22 in each phenotype. Finally, 102 participants were analyzed. Of these, 74 subjects were stratified into three groups: CSP_{2D} group, skeletal Class II hyperdivergent patients without TSD ($n = 23$); CSP_{3D} group, skeletal Class II hyperdivergent patients with TSD ($n = 25$); and non-CSP group, skeletal Class I normodivergent patients without TSD ($n = 26$).

The study was approved by the Institutional Review Board of Kyung Hee University Dental Hospital (KH-DT 22015) and was performed in accordance with the Declaration of Helsinki on Medical Research. All participants signed an informed consent form before registration in the institutional database, which included their approval for use for research purposes.

Polysomnography and questionnaire

Full standard in-laboratory polysomnography was analyzed by one trained sleep technologist following the standard scoring guideline from the 2012 American Academy of Sleep Medicine manual [25]: overall apnea–hypopnea index (AHI), apnea index (AI), hypopnea index (HI), AHI in a supine position (AHI_{SUP}), AHI in a rapid eye movement (REM) stage (AHI_{REM}), AHI in a non-REM stage (AHI_{NREM}), respiratory disturbance index (RDI), oxygen desaturation index (ODI), the lowest oxygen saturation ($LSaO_2$), arousal index (ArI), and wake period after sleep onset (WASO). Body mass index (BMI; kg/m^2) was calculated to determine its impact on obesity. The representative OSA symptoms of excessive daytime sleepiness were assessed using ESS.

3D skeletal pattern classification

CBCT scans were acquired using an Alphard-Vega 3030 machine (Asahi Roentgen Ind., Co. Ltd., Kyoto, Japan) based on the following parameters: 8 mA, 80 kV, and a scan time of 17 s. The field of view was $15.4 \times 15.4\ cm^2$ with an isotropic voxel unit of $0.3\ mm^3$ to contain the whole UA image from the nares to the epiglottis base. CBCT images taken in standardized head, neck, and jaw postures at the end of expiration within a respiratory cycle were collected from the ZeTTA PACS (picture archiving and communication systems, TaeYoung Soft, Anyang, Korea) database. Data were

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exported in a Digital Imaging and Communications in Medicine (DICOM) format.

3D skeletal analysis for pattern classification was performed using an *in vivo* 3D imaging software (Anatomage, Inc., Santa Clara, CA, USA) following the head reorientation process. All landmarks, reference planes, and measurements are described in Fig. 1 and Table 1. The sagittal skeletal pattern was classified by ANB angle: Class I, 0°<ANB < 4°; and Class II, ANB ≥ 4°. The vertical pattern was classified by mandibular plane angle to FH plane (MP-FH): normodivergent, 25°≤MP-FH < 30°; hyperdivergent, MP-FH ≥ 30°; and hypodivergent pattern, MP-FH < 25° (Fig. 1A). To categorize the transverse skeletal pattern, the transverse discrepancy index (TDI) was calculated as the maxillary alveolar width (MxAW) minus the mandibular alveolar width (MnAW), which was measured on the coronal sectional image at the level of the center of resistance of the first molars (Fig. 1B). TSD was defined as a TDI < −2.5 mm, while non-TSD was defined as a TDI > −2 mm. Additionally, the palatal height index (PHI) and nasal cavity width ratio (NCWR) were calculated to support the TSD.

3D UA segmentation and measurement

UA volume mask was specified and semi-automatically delineated through a region-growing process using the Mimics software (version 20.0, Materialise N.V., Leuven, Belgium), and divided into nasal airway volume (NAV) and pharyngeal airway volume (PAV). Twelve landmarks were manually digitized to generate six reference planes: the Na–Ba plane, ANS–PNS

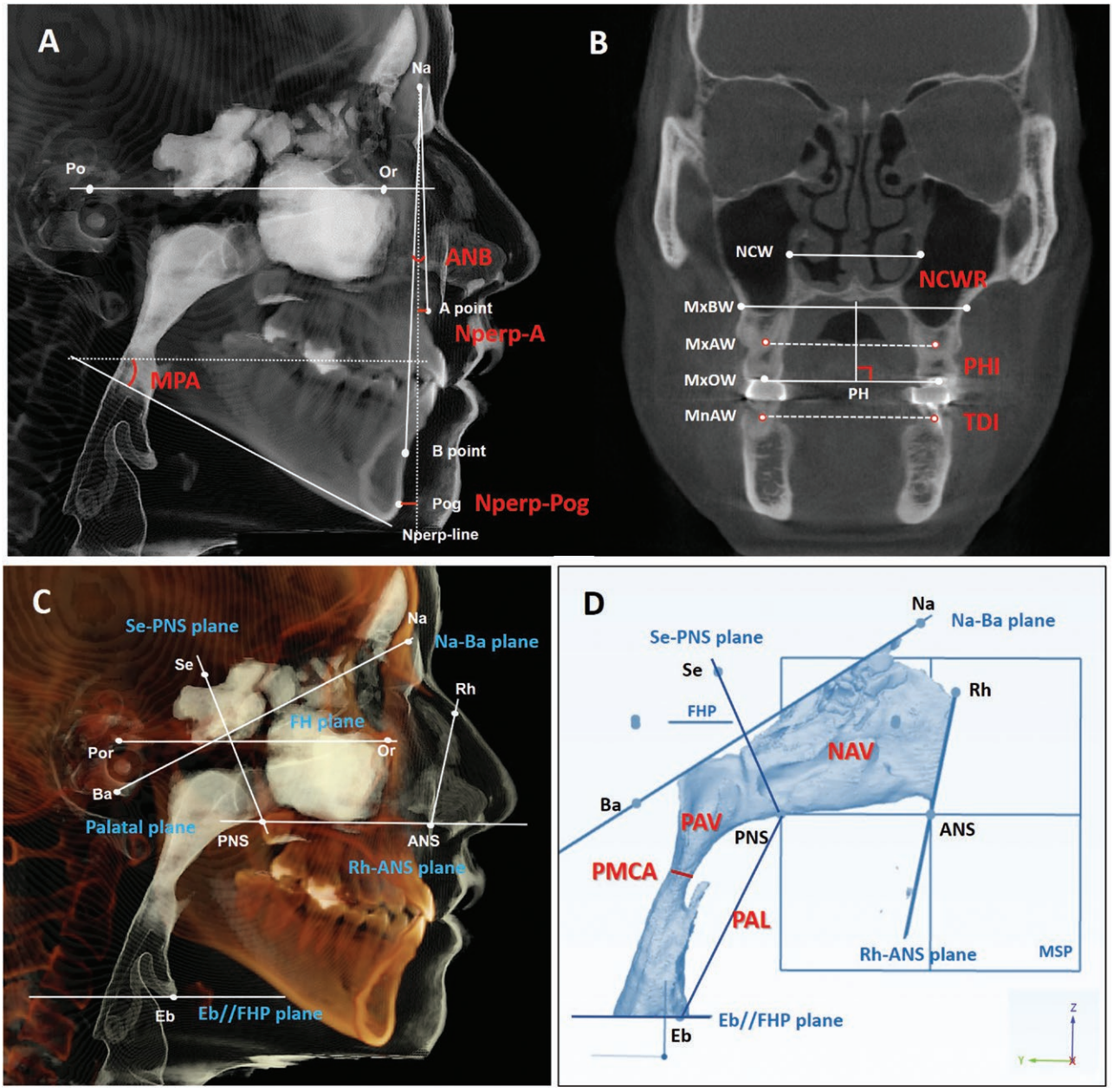


Figure 1. 3D morphometric analysis for skeletal pattern classifications and regional upper airway measurements on the CBCT images. A, Sagittal and vertical skeletal variables. B, Transverse skeletal variables. C, Six reference planes reconstructed on the 3D volume image to delineate nasal and pharyngeal airway volume masks. D, Upper airway measurements on the segmented volume mask. Abbreviations and definitions are described in Table 1.

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Table 1. Definition of craniofacial skeletal and upper airway variables measured on CBCT images.

Variables	Definition
Skeletal variables for 3D pattern classification	
ANB (°)	The angle between point A and point B at nasion (N) to represent sagittal maxillomandibular relationship
Nperp_A (mm)	The shortest distance from point A to N-perpendicular line
Nperp_Pog(mm)	The shortest distance from Pogonion to N-perpendicular line
MP-FH (°)	Mandibular plane angle to Frankfurt horizontal plane
NCWR (%)	Nasal cavity width ratio, the widest distance of the nasal cavity (NCW) divided by maxillary basal width (MxBw) × 100
TDI (mm)	Transverse discrepancy index, maxillary alveolar width (MxAW) subtracted by mandibular alveolar width (MnAW)
PHI (%)	Palatal height index, palatal height (PH) divided by maxillary occlusal width (MxOW) × 100
Upper airway variables	
NAV ($mm^{3)}$)	Nasal airway volume, semi-automatically measured nasal airway volume mask delineated by Na–Ba plane, Se–PNS plane, Rh–ANS plane, and ANS–PNS plane
PAV ($mm^{3)}$)	Pharyngeal airway volume, semi-automatically measured pharyngeal airway volume mask delineated by Se–PNS plane, epiglottis base (Eb)-plane, and pharyngeal walls
PMCA ($mm^{2)}$)	Pharyngeal minimal cross-sectional area automatically captured within pharyngeal volume mask
PAL (mm)	Pharyngeal airway length, distance from PNS to Eb on the midsagittal sectional image

The relevant landmarks, reference planes, and measurements are described in Fig. 1.

plane, Se–PNS plane, Rh–ANS plane, FH plane, and Eb//FHP plane for setting the primary UA volume of interest (Figure 1C). The mask contour was edited through a manual thresholding process to remove noise, artifacts, and irrelevant parts: from –400 to –500 Hounsfield units (HU) for the NAV and from –1200 to –400 HU for the PAV. The volume masks were exported in a standard tessellation language (STL) file format to the 3-Matic Medical Software (version 12.0, Materialise N.V., Leuven, Belgium). The final NAV mask was created after removing the adjacent sinusoidal volumes. The Na–Ba plane, ANS–PNS plane, Se–PNS plane, and Rh–ANS plane served as the uppermost, lowermost, rearmost, and frontline boundaries of the final NAV, respectively. The PAV mask was generated using the Se–PNS and Eb-planes as the uppermost and lowermost borderlines, respectively. Additionally, the pharyngeal minimal cross-sectional area (PMCA) and pharyngeal airway length (PAL) were measured using an *in vivo* 3D imaging software (Anatomage, Inc., Santa Clara, CA).

All CBCT measurements were obtained twice by an orthodontic specialist at 2-week intervals, and the mean values of the two measurements were used for the analysis. Intraclass correlation coefficients (ICCs) used to assess the intraobserver reliability for skeletal and UA measurements ranged from 0.975 to 0.996 and from 0.935 to 0.990, respectively.

Statistical analysis

The normality distribution of each variable was assessed using the Shapiro–Wilk test. Intergroup comparisons were conducted using an independent *t*-test or analysis of variance (ANOVA) with a Bonferroni post hoc comparison. Gender percentages were compared between groups using the chi-square test. Based on the results of Pearson's correlation analysis between CBCT morphometric and polysomnographic variables, multiple linear regression analysis was performed to identify any CBCT variables affecting OSA severity, after adjusting for gender, age, and BMI. The probability of multiperspective characteristics among CSP_{2D}, CSP_{3D}, and non-CSP groups was introduced to represent the results of ANOVA. All analyses were performed using SPSS version 20 software (SPSS Inc., Chicago, IL, USA). A *P*-value of <.05 was considered significant.

Results

The baseline characteristics of the participants are described in Table 2. Male patients who were more obese (BMI, $P < .01$) presented with greater OSA severity than female patients as indicated by six polysomnographic variables, although they showed smaller skeletal discrepancies than women especially in the sagittal (ANB and Nperp_Pog, $P < .05$) and vertical (MP-FH, $P < .05$) dimensions with greater PAV ($P < .05$) and PAL ($P < .001$). No gender differences were observed in the mean age, transverse aspect of skeletal discrepancy (TDI, PHI, and NCWR), NAV, and pharyngeal MCA.

The results of inter-variable regression analysis to estimate the influence of five selected CBCT variables (ANB, MP-FH, TDI, PMCA, and PAL) from the prior correlation analysis (Supplemental Table 1) on the multiple polysomnographic variables are described in Table 3. ANB and TDI proved to be significant craniofacial predictors for each of AHI, AHI_{SUP}, AI, RDI, ODI, and $LSaO_2$, with no evidence of multicollinearity among the independent variables.

Skeletal inter-class comparisons of OSA severity showed significant differences in each of 3 dimensions (Table 4). Skeletal Class II samples with retrusive mandibles exhibited more severe OSA than Class I samples in five polysomnographic variables, despite their younger age. Hyperdivergent samples showed greater OSA severity in four polysomnographic variables than normodivergent samples despite their younger age and lower obesity, while displaying no significant difference from hypodivergent samples having greater obesity. TSD samples presented more severe OSA than non-TSD samples in six polysomnographic variables, with no differences in mean age and BMI. The UA variable that was commonly susceptible to skeletal restrictions in all dimensions was PMCA.

As a result of the inter-phenotype comparison, the CSP group represented greater AHI, AHI_{SUP}, AHI_{NREM}, $AHI_{REM/NREM}$, AI, RDI, ODI, and $LSaO_2$, with smaller NAV, smaller PMCA, and longer PAL than the non-CSP group, despite younger age and lower obesity (Table 5). Intragroup comparison of the CSP indicated that the CSP_{3D} subgroup (52.1%) presented with far more severe OSA than the age- and BMI-matched CSP_{2D} subgroup (47.9%). The contrast in the probability of multiperspective characteristics among the CSP_{3D}, CSP_{2D}, and non-CSP groups is illustrated in Fig. 2, based on the result of ANOVA. To compensate for scale differences, all variables have been normalized using the following formula: (*X*–minimum value)/(maximum value–minimum value).

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Table 2. Demographic, polysomnographic, and CBCT morphometric characteristics, representing gender-difference.

Variables			Total (n = 102)	Male (n = 71)	Female (n = 31)	P-value*
Demographic		Age (year)	44.4 ± 14.1	43.5 ± 13.2	47.4 ± 13.6	.052
		BMI (kg/m^2)	24.7 ± 3.8	25.4 ± 3.8	23.2 ± 3.4	.006**
Polysomnographic		AHI (event/h)	35.9 ± 20.3	40.3 ± 21.5	25.5 ± 12.0	.001**
		AHI_{SUP} (event/h)	46.4 ± 23.0	51.0 ± 23.7	35.6 ± 17.1	.002**
		AI (event/h)	13.4 ± 18.6	16.6 ± 21.2	5.9 ± 5.0	.008**
		HI (event/h)	22.6 ± 13.2	23.7 ± 14.1	20.0 ± 10.8	.194
		RDI (event/h)	40.0 ± 19.3	44.0 ± 20.8	30.8 ± 10.4	.001**
		ODI (event/h)	23.5 ± 18.8	28.2 ± 19.9	12.6 ± 9.4	.000***
		$LSaO_2$ (%)	83.3 ± 6.3	82.2 ± 6.3	85.7 ± 5.4	.008**
3D morphometric	Skeletal	ANB (°)	5.0 ± 2.7	3.8 ± 2.7	5.6 ± 2.8	.032*
		Nperp_A (mm)	0.7 ± 1.9	0.7 ± 1.9	0.7 ± 2.0	.962
		Nperp_Pog (mm)	−7.6 ± 6.1	−6.6 ± 4.4	−8.6 ± 4.4	.041*
		MP-FH (°)	29.4 ± 5.8	28.1 ± 5.8	31.7 ± 5.5	.014*
		TDI (mm)	−1.6 ± 2.1	−1.7 ± 2.2	−1.5 ± 1.9	.671
		PHI (%)	43.4 ± 4.2	42.9 ± 4.1	44.6 ± 4.1	.065
		MxBW (mm)	66.7 ± 4.4	67.5 ± 4.4	64.8 ± 3.6	.005**
		NCW (mm)	32.5 ± 3.9	32.8 ± 4.0	31.9 ± 3.7	.272
		NCWR (%)	48.9 ± 6.4	48.9 ± 5.9	49.1 ± 7.3	.891
	Upper Airway	NAV (mm^3)	14 535.1 ± 5214.1	14 422.2 ± 5561.1	14 798.6 ± 4373.2	.743
		PAV (mm^3)	26 415.4 ± 9004.1	27 601.5 ± 9160.2	24 607.2 ± 9310.0	.020*
		PMCA (mm^2)	128.0 ± 66.0	133.8 ± 71.1	114.7 ± 50.7	.186
		PAL (mm)	71.1 ± 6.0	73.1 ± 5.3	66.4 ± 4.9	.000***

Analyzed by independent t-test.
*$P < 0.05$.
**$P < 0.01$.
***$P < 0.001$.
Abbreviations: AHI, apnea–hypopnea index; AHI_{SUP}, AHI measured at supine position; AI, apnea index; BMI, body mass index; HI, hypopnea index; $LSaO_2$, the lowest oxygen saturation; MPA, mandibular plane angle; MxBW, maxillary basal width; NAV, nasal airway volume; NCW, nasal cavity width; NCWR, nasal cavity width ratio; Nperp_A, nasion-perpendicular to point A; Nperp_Pog, nasion-perpendicular to Pognion; ODI, oxygen desaturation index; PAL, pharyngeal airway length; PAV, pharyngeal airway volume; PHI, palatal height index; PMCA, pharyngeal minimal cross-section area; RDI, respiratory disturbance index; TDI, transverse discrepancy index.

Discussion

The present study first identified and characterized the CSP of OSA patients from a multidimensional perspective, in contrast to previous studies that focused on individual craniofacial variables or compared sagittal and vertical skeletal discrepancies [4, 10–12, 26]. Specifically, this study explored the significance of TSD on OSA manifestations to address the comprehensive 3D skeletal attributes in the CSP, based on multivariable polysomnographic investigation. We discovered that the CSP of OSA patients had greater disease severity than the non-CSP of OSA patients, and the OSA severity was increased when TSD was accompanied with sagittal and vertical skeletal discrepancies in the CSP_{3D} patients (Table 5). This provides clinical evidence that recognition of the CSP should be a crucial step in the management of OSA, advocating for tailored craniofacial treatment rather than the application of nonspecific symptomatic treatments [1, 2].

Previously, most studies attempted to identify craniofacial predictors for OSA severity on 1- or 2-dimensional CBCT sections [6, 8, 11–14]. However, the significant impact of any single craniofacial variable on a single polysomnographic variable remains controversial across studies, because of variations in sample sizes, demographics, imaging techniques and conditions, definitions of upper airway measures, and the range of polysomnographic variables [27]. Wang *et al.* [11] reported that the craniofacial predictors in 95 Chinese patients (female fraction, 14.7%; mean age, 36.0; BMI, 26.1; AHI, 36.3) were mandibular body length and retropalatal airway length. Although, they found that CBCT served as a sectional determinant of AHI alone, with no inclusion of the transverse dimension. By contrast, Gurgel *et al.* [12] found that frontomaxillary suture width and frontal ramus angle significantly affected the OSA severity in 20 Brazilian patients (female fraction, 55.0%; mean age, 48.4; BMI, 27.1), without integration with sagittal and vertical skeletal features. Although their study highlighted the significance of the transverse contribution to OSA, their outcome variables were irrelevant to skeletal pattern classification which is crucial for target-based skeletal management. The craniofacial predictors obtained from the present study with 102 Korean OSA patients (female fraction, 30.4%; mean age, 44.4; BMI, 24.7; AHI, 35.9) were ANB, TDI, and PMCA. These predictors could be obtained from the correlation between skeletal variables optimized for 3D pattern classifications and multiple polysomnographic variables. Nonetheless, this variable-to-variable relationship may not translate into a comprehensive disease severity predictor due to individual diversity even within the same ethnicity and gender.

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Table 3. Skeletal and upper airway variables affecting the OSA severity, when controlled by gender, age, and BMI.

Dependent variables	Independent variables	*B* estimate	SE	*β*	*t*	*P*-value	VIF	Adjusted R^2
AHI	Constant	–13.59	28.54		–0.48	.635		0.40
	ANB	2.37	0.82	0.32	2.88	.005**	2.01	
	MP-FH	0.16	0.38	0.05	0.41	.680	1.95	
	TDI	–2.66	0.83	–0.27	–3.21	.002**	1.19	
	PMCA	–0.06	0.03	–0.21	–2.09	.039*	1.65	
	PAL	–0.12	0.33	–0.03	–0.35	.727	1.55	
AHI_{SUP}	Constant	–42.28	31.68		–1.33	.185		0.42
	ANB	2.48	0.91	0.29	2.72	.008**	2.01	
	MP-FH	0.58	0.43	0.15	1.37	.175	1.95	
	TDI	–2.98	0.92	–0.27	–3.25	.002**	1.19	
	PMCA	–0.07	0.03	–0.20	–2.04	.044*	1.65	
	PAL	0.13	0.37	0.03	0.36	.722	1.55	
AI	Constant	–64.00	27.41		–2.33	.022		0.36
	ANB	1.40	0.79	0.21	1.77	.049*	2.01	
	MP-FH	0.26	0.37	0.08	0.70	.485	1.95	
	TDI	–2.72	0.79	–0.31	–3.42	.001**	1.19	
	PMCA	–0.06	0.03	–0.21	–2.04	.055	1.65	
	PAL	0.53	0.32	0.17	1.65	.102	1.55	
RDI	Constant	–2.93	27.38		–0.11	.915		0.39
	ANB	2.15	0.79	0.30	2.73	.008**	2.01	
	MP-FH	0.15	0.37	0.04	0.39	.694	1.95	
	TDI	–2.50	0.79	–0.27	–3.15	.002**	1.19	
	PMCA	–0.07	0.03	–0.23	–2.25	.027*	1.65	
	PAL	–0.15	0.32	–0.05	–0.46	.645	1.55	
ODI	Constant	–46.86	27.13		–1.73	.088		0.37
	ANB	1.77	0.78	0.26	2.27	.026*	2.01	
	MP-FH	–0.04	0.37	–0.01	–0.11	.916	1.95	
	TDI	–2.41	0.79	–0.27	–3.07	.003**	1.19	
	PMCA	–0.03	0.03	–0.10	–0.95	.347	1.65	
	PAL	0.16	0.31	0.05	0.50	.616	1.55	
$LSaO_2$	Constant	100.17	9.29		10.78	.000		0.33
	ANB	–0.62	0.27	–0.27	–2.32	.022*	2.01	
	MP-FH	0.15	0.12	0.14	1.20	.232	1.95	
	TDI	1.32	0.27	0.44	4.92	.000***	1.19	
	PMCA	0.01	0.01	0.12	1.15	.254	1.65	
	PAL	–0.10	0.11	–0.10	–0.93	.356	1.55	

Tested by multiple linear regression analysis, adjusted by gender, age, and BMI. *P <.05; **P <.01; ***P <.001.
Abbreviations are described in Table 2.

One step further, we performed skeletal inter-class comparisons in each of the 3D assuming a predominance of any skeletal class contributing to OSA. Notably, the vertical skeletal vulnerability of the hyperdivergent pattern displayed greater disease severity than the normodivergent patients who were older and more obese, although a significant relationship between MP-FH and polysomnographic variables was not found in the regression analysis. This inconsistency might be attributed to the sample characteristics of hyperdivergent patients who manifested greater Class II sagittal discrepancy (ANB, $P < .001$; Nperp_Pog, $P < .001$) and greater TSD (TDI, $P < .05$; PHI, $P < .001$) than the normodivergent patients on average. Likewise, this complexity observed in an individual patient can hardly be explained by single-dimensional analysis. A recent meta-analysis by Neelapu *et al.* [4] supported strong evidence on the correlation between OSA and vertical deformity compared with sagittal deformity, however, they pointed out larger heterogeneity across studies regarding the influence of skeletal deformity. Interestingly, the significance of the inter-class differences in OSA severity was the most remarkable in the transverse aspect (Table 4). When TSD and non-TSD samples were age- and BMI-matched, without significant gender difference, the TSD group seemed to represent more severe OSA. However, the effect of TSD could not be clearly discerned because the TSD samples had hyperdivergent patterns with retrusive chins, practically presenting with 3D skeletal restriction.

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Table 4. Comparison of all variables according to skeletal classifications in three-dimensional aspects.

Variables	Sagittal classification*			Vertical classification†				Transverse classification*		
	Class I (n = 40)	Class II (n = 62)	P-value	Normodivergent (a, n = 25)	Hypodivergent (b, n = 23)	Hyperdivergent (c, n = 54)	P-value (post hoc)	N_TSD (n = 54)	TSD (n = 35)	P-value*
Gender (male %)	71.7	68.9	.702	62.5	82.6	67.9	ns	60.7	70.5	.530
Age (year)	48.9 ± 11.4	41.5 ± 15.0	.009**	49.1 ± 12.8	46.2 ± 12.8	41.5 ± 14.7	$a > c^{\dagger}$	45.9 ± 14.2	41.9 ± 13.7	.172
BMI (kg/m^2)	25.5 ± 3.6	24.2 ± 3.1	.079	26.2 ± 4.0	25.7 ± 3.7	23.6 ± 3.4	$a,b > c^{\dagger\dagger}$	24.6 ± 3.8	24.9 ± 3.7	.724
AHI (event/h)	28.9 ± 14.8	40.4 ± 22.1	.005**	25.7 ± 10.7	35.9 ± 17.9	40.5 ± 23.0	$a < c^{\dagger}$	28.7 ± 14.4	47.1 ± 23.1	.000***
AHI_{SUP} (event/h)	38.5 ± 20.6	51.5 ± 23.2	.005**	33.3 ± 14.8	44.6 ± 20.6	53.1 ± 24.6	$a < c^{\dagger\dagger}$	37.4 ± 18.4	60.6 ± 22.4	.000***
AI (event/h)	7.6 ± 9.2	17.1 ± 21.9	.012*	7.5 ± 8.2	9.2 ± 12.9	17.9 ± 22.6	$a < c^{\dagger}$	6.5 ± 7.1	24.1 ± 15.0	.000***
HI (event/h)	21.3 ± 12.1	23.4 ± 14.0	.441	18.3 ± 8.8	26.7 ± 15.5	22.8 ± 13.5	ns	22.1 ± 11.9	23.3 ± 15.2	.670
RDI (event/h)	33.4 ± 14.4	44.3 ± 20.8	.005**	30.8 ± 10.9	40.2 ± 17.3	44.2 ± 21.7	$a < c^{\dagger}$	33.4 ± 13.8	50.3 ± 20.0	.000***
ODI (event/h)	17.9 ± 11.8	27.1 ± 21.5	.016*	16.2 ± 7.9	22.8 ± 17.1	27.2 ± 22.0	ns	18.4 ± 13.7	31.6 ± 18.7	.000***
$LSaO_2$ (%)	84.9 ± 4.9	82.2 ± 6.8	.058	84.3 ± 4.7	84.5 ± 5.7	82.2 ± 7.0	ns	85.3 ± 4.8	79.9 ± 6.9	.000***
ANB (°)	2.2 ± 1.0	6.7 ± 1.0	.000***	3.3 ± 2.3	3.4 ± 2.1	6.3 ± 2.4	$a, b < c^{\dagger\dagger\dagger}$	4.5 ± 2.7	5.5 ± 2.6	.079
Nperp_A (mm)	0.1 ± 1.9	1.0 ± 1.8	.024*	1.0 ± 2.2	0.4 ± 1.5	0.6 ± 1.9	ns	0.9 ± 1.8	0.3 ± 1.9	.127
Nperp_Pog (mm)	−2.7 ± 2.3	−10.8 ± 5.6	.000***	−3.5 ± 3.7	−4.2 ± 3.1	−10.9 ± 6.0	$a, b > c^{\dagger\dagger\dagger}$	−6.2 ± 5.3	−9.8 ± 6.6	.004*
MP-FH (°)	26.1 ± 4.1	32.0 ± 5.4	.000***	27.0 ± 1.5	22.4 ± 1.5	34.1 ± 3.7	$b < a < c^{\dagger\dagger\dagger}$	28.6 ± 5.5	31.4 ± 5.7	.018*
TDI (mm)	−1.1 ± 1.8	−1.9 ± 2.2	.050	−1.1 ± 1.7	−0.8 ± 2.4	−2.1 ± 2.0	$b > c^{\dagger}$	−0.4 ± 1.5	−3.8 ± 1.3	.000***
PHI (%)	42.2 ± 4.1	44.1 ± 4.1	.026*	42.5 ± 2.9	40.7 ± 4.1	45.0 ± 4.0	$a,b < c^{\dagger\dagger\dagger}$	42.6 ± 3.9	44.7 ± 4.3	.012*
MxBW (mm)	67.7 ± 4.2	66.0 ± 4.4	.067	67.1 ± 3.7	68.1 ± 3.6	65.9 ± 4.8	ns	67.7 ± 4.1	65.1 ± 4.3	.004**
NCW (mm)	33.2 ± 4.0	32.1 ± 3.8	.157	33.3 ± 4.5	32.6 ± 3.5	32.2 ± 3.8	ns	32.9 ± 4.0	31.9 ± 3.6	.188
NCWR (%)	49.3 ± 7.1	48.7 ± 5.8	.640	49.6 ± 7.0	49.0 ± 6.2	48.1 ± 6.2	ns	48.7 ± 6.5	49.1 ± 6.1	.740
NAV (mm^3)	15 699.1 ± 4828.8	13 791.0 ± 5352.2	.074	15 860.8 ± 3795.4	15 703.4 ± 7156.3	13 427.8 ± 4588.6	ns	15 563.9 ± 5205.5	12 925.9 ± 4866.7	.013*
PAV (mm^3)	28 027.8 ± 9160.2	26 027.8 ± 9005.4	.068	27 852.2 ± 8890.3	27 052.0 ± 9320.5	26 627.8 ± 9100.3	ns	27 565.8 ± 9300.5	25 990 ± 9530.3	.089
PMCA (mm^2)	160.7 ± 67.7	107.1 ± 5.0	.000***	147.4 ± 62.7	146.0 ± 70.3	111.4 ± 62.2	$a,b > c^{\dagger}$	140.5 ± 68.4	108.5 ± 57.7	.017*
PAL (mm)	70.0 ± 56.1	71.7 ± 6.4	.159	70.9 ± 6.1	69.2 ± 4.6	72.0 ± 6.3	ns	70.7 ± 5.8	72.3 ± 5.6	.053

Tested by independent t-test (*) or ANOVA (†) with Bonferroni post hoc comparison. * or † $P < .05$; ** or †† $P < .01$; *** or ††† $P < .001$. Abbreviations are described in Table 2.

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Table 5. Characteristics of the craniofacial skeletal phenotype of OSA in comparison with the non-craniofacial skeletal phenotype of OSA.

Variables	non-CSP ($n = 26$)	Craniofacial skeletal phenotype (CSP)				*P*-value (non-CSP vs. CSP)
		CSP_Total ($n = 48$)	CSP_{2D} ($n = 23$)	CSP_{3D} ($n = 25$)	*P*-value (CSP_{2D} vs. CSP_{3D})	
Gender (male %)	69.2	67.4	56.6	69.0	.039†	.872
Age (year)	47.2 ± 1.8	39.8 ± 1.9	39.4 ± 14.9	41.0 ± 15.5	.703	.028*
BMI (kg/m^2)	26.9 ± 4.0	23.6 ± 3.4	22.2 ± 2.5	24.1 ± 3.9	.058	.008**
AHI (event/h)	29.6 ± 16.1	41.3 ± 23.9	29.3 ± 14.2	51.5 ± 26.3	.001**	.001**
AHI_{SUP} (event/h)	37.2 ± 18.7	53.0 ± 24.0	39.4 ± 18.1	64.4 ± 22.5	0.000***	.000***
AI (event/h)	7.5 ± 10.1	19.7 ± 23.7	8.2 ± 5.3	31.9 ± 28.0	0.000***	.000***
HI (event/h)	22.1 ± 11.7	21.9 ± 13.1	23.1 ± 12.5	20.7 ± 15.1	.577	.742
RDI (event/h)	34.0 ± 16.2	45.1 ± 22.3	34.3 ± 13.2	54.0 ± 25.0	.002**	.001***
ODI (event/h)	19.7 ± 13.3	28.3 ± 23.0	17.6 ± 13.5	37.0 ± 26.3	.003**	.003***
$LSaO_2$ (%)	83.9 ± 4.9	81.5 ± 7.1	86.4 ± 4.2	77.4 ± 6.5	0.000***	.001**
ArI (event/hr)	29.3 ± 13.9	37.6 ± 19.2	29.5 ± 14.8	44.4 ± 20.0	.006**	.056
WASO (min)	53.4 ± 37.6	44.74 ± 35.5	51.1 ± 38.7	39.4 ± 32.3	.268	.331
AHI_{REM} (event/h)	32.7 ± 16.7	34.4 ± 20.2	29.0 ± 12.9	39.0 ± 24.1	.095	.712
AHI_{NREM} (event/h)	21.1 ± 14.9	40.5 ± 27.4	25.5 ± 16.6	53.1 ± 28.6	0.000***	.000***
$AHI_{REM/NREM}$	1.9 ± 1.0	1.1 ± 0.7	1.5 ± 0.8	0.8 ± 0.4	.001**	.000***
ESS	8.3 ± 4.0	10.2 ± 4.8	9.3 ± 5.2	11.0 ± 4.4	.253	.095
ANB (°)	2.1 ± 1.0	7.0 ± 1.9	7.2 ± 1.9	6.9 ± 2.0	.333	.000***
Nperp_A (mm)	0.3 ± 1.7	0.8 ± 1.8	1.2 ± 1.6	0.6 ± 1.9	.287	.607
Nperp_Pog (mm)	–2.4 ± 2.8	–11.9 ± 5.8	–11.6 ± 4.6	–12.1 ± 7.2	.769	.000***
MP-FH (°)	24.3 ± 3.2	34.4 ± 3.8	34.5 ± 3.3	34.2 ± 4.5	.904	.000***
TDI (mm)	–0.2 ± 1.6	–2.3 ± 2.1	–0.6 ± 1.3	–3.6 ± 1.4	0.000***	.000***
PHI (%)	40.2 ± 3.0	44.9 ± 4.2	44.1 ± 4.1	45.2 ± 4.6	.259	.000***
MxBW (mm)	68.3 ± 3.8	65.4 ± 4.4	66.4 ± 4.3	65.3 ± 4.4	.167	.027*
NCW (mm)	35.3 ± 3.9	31.7 ± 3.8	31.4 ± 3.8	31.9 ± 4.0	.866	.199
NCWR (%)	48.9 ± 7.3	48.6 ± 5.9	47.4 ± 5.8	49.6 ± 6.0	.359	.855
NAV (mm^3)	15 796.8 ± 5549.0	12 815.4 ± 4301.1	14 400.7 ± 3900.1	11 023.7 ± 3523.0	.022*	.025*
PAV (mm^3)	28 600.7 ± 9228.2	25 003.7 ± 9228.2	25 803.2 ± 8990.3	24 520.6 ± 8828.3	.369	.077
PMCA (mm^2)	165.5 ± 67.8	101.1 ± 38.7	125.5 ± 45.7	89.6 ± 32.7	.033*	.000***
PAL (mm)	68.3 ± 4.0	72.3 ± 5.3	70.7 ± 5.1	74.2 ± 5.0	.032*	.001**

Tested by independent *t*-test (*) or chi-square test (†). $^{*}P < .05$; $^{**}P < .01$; $^{***}P < .001$.
Abbreviations: AHI_{NREM}: AHI during non-REM sleep; AHI_{REM}: AHI during REM sleep; $AHI_{REM/NREM}$: the ratio of AHI_{REM} to AHI_{NREM}; ArI: arousal index; ESS: Epworth sleepiness scale; WASO: wake after sleep onset. Other abbreviations are described in Table 1.

Accordingly, the main originality of this study is that a subject-to-subject comparison of OSA presentations according to the OSA phenotype was designed to reach the practical goal of identifying CSP when diagnosing the OSA population. The CSP patients showed worse values in obstructive and oxygen-related indices than the non-CSP patients despite being younger and less obese (Table 5). This finding is in contrast to that of our previous 2D study [26], which reported that the Class II hyperdivergent patterns were not correlated with OSA severity or sleepiness symptoms, possibly due to the absence of a properly established control group in that study. The present study adopted non-CSP of OSA patients as the control group, instead of healthy individuals, aiming to provide criteria to distinguish the CSP from non-CSP in the diagnosis of OSA patients. Furthermore, the CSP_{3D} patients exhibited more severe OSA and more vulnerable nasal and pharyngeal airways than the CSP_{2D} patients. This could imply a significant contribution of TSD to OSA severity because the mean age, BMI, and sagittal and vertical skeletal patterns were well matched. As visualized by a probability comparison of multidimensional variables among groups (Fig. 2), the CSP_{3D} additionally represented a higher arousal index and lower $AHI_{REM/NREM}$ than the CSP_{2D}, indicating worse sleep quality. This study emphasizes the importance of transverse maxillary assessment encompassing the nasal cavity, not to be disregarded only through the cephalometric diagnosis.

The significance of craniofacial treatment for adult OSA patients has been underrated as an adjunctive or last-priority option in the field of sleep medicine, due to the lack of a clear demonstration of the CSP of OSA [1–3]. Sutherland *et al.* [27, 28] highlighted the need for craniofacial phenotyping in OSA patients using facial photographs. Despite the simplicity and predictability of OSA in some cases especially with regional adiposity, it is unclear whether surface facial measurements reflect the underlying skeletal risk of UA collapsibility. Ng *et al.* [29] suggested a craniofacial phenotype of OSA using a single CBCT metric of maxillomandibular volume (MMV) to determine the therapeutic effects of

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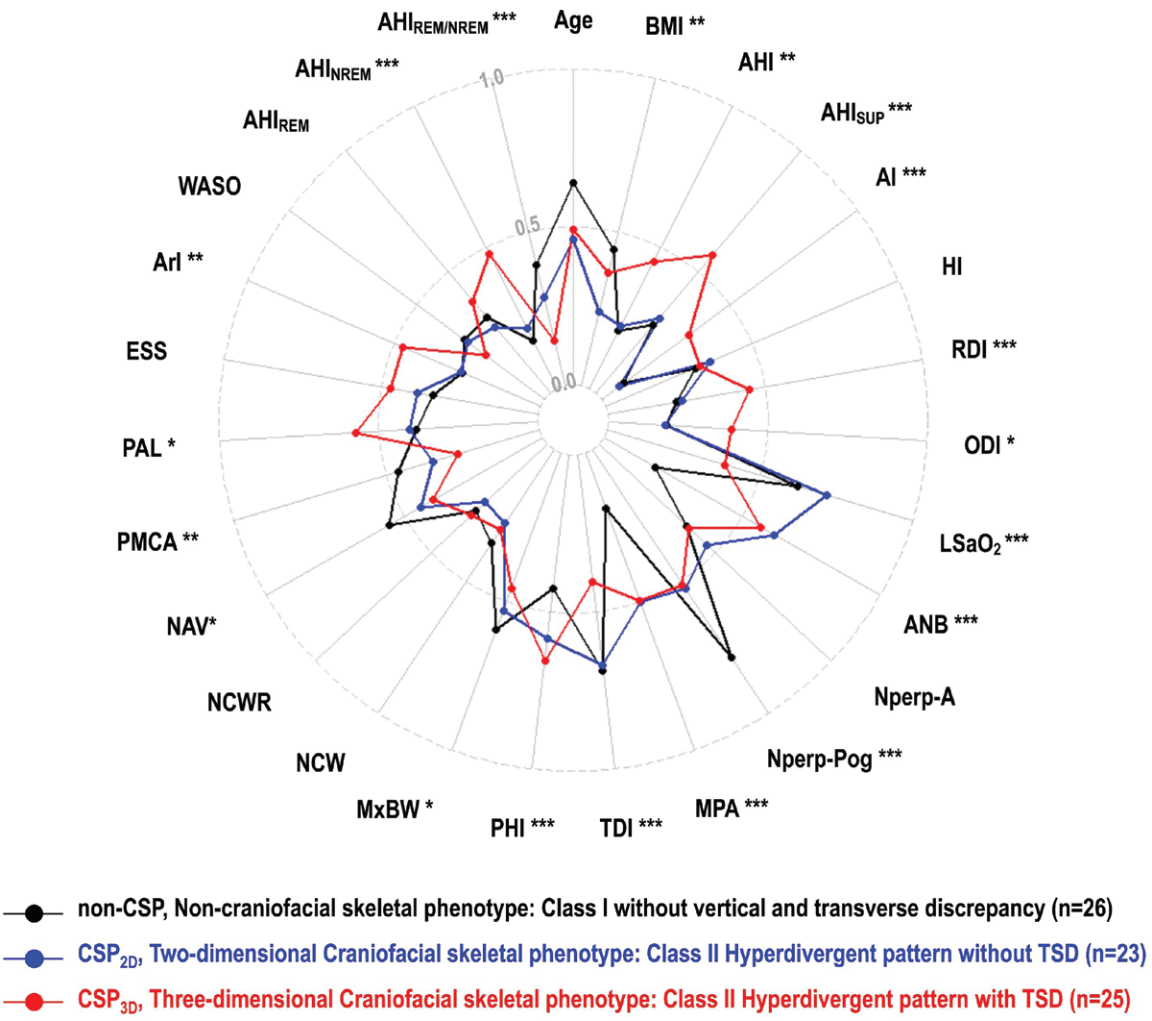


Figure 2. A normalized probability diagram comparing 3D craniofacial patterns, airway dimensions, and various polysomnographic variables among CSP2D, CSP3D, and non-CSP groups, with ANOVA testing and significance levels indicated (*$P < .05$; **$P < .01$; ***$P < .001$). Abbreviations are described in Table 5.

weight loss and positive airway pressure on OSA when stratified by craniofacial phenotype. Although they approached skeletal measurement, MMV indicating a geometric maxillomandibular enclosure neither represented the actual skeletal volume nor accounted for internal skeletal-restrictive features. Considering that craniofacial phenotyping aims to identify the main skeletal predisposition to UA collapsibility, internal skeletal and UA risks should be included. In this context, the present study first defined the craniofacial skeletal phenotype (CSP) and characterized it in both 2D and 3D for orthodontists to take the initiative in phenotype-based treatment planning for OSA.

Another originality is whole UA image including nasal and pharyngeal airways was analyzed to find any dimensional vulnerability of UA in the CSP patients in relation to skeletal restriction, based on the digitally-created segmentation model analysis. Makkonen *et al.* [14] reported that the nasal MCA on each CBCT section was not associated with OSA severity, supporting the clinical consensus that NAV alone is an insufficient variable to predispose OSA pathogenesis [21, 30]. Interestingly, however, significant differences were found in NAV as well as PMCA and PAL between the CSP and non-CSP groups, and between the CSP_{2D} and CSP_{3D} subgroups, supposedly due to the impact of TSD with nasomaxillary constriction, although NAV showed no correlation with any single polysomnographic variable independently. In contrast, PAV showed no significance in predicting OSA, corresponding to Buchanan *et al.*'s study [31]. This is because the computed PAV value, which is proportional to both PMCA and PAL, may be compensated by small PMCA and large PAL numbers in the CSP of OSA patients [10, 11]. Rather than PAV, therefore, the PMCA and PAL should be primarily noted in the CSP patients, although the pharyngeal dimension cannot always account for pharyngeal collapsibility.

In our single-ethnicity population study, we controlled for age, gender, and obesity as confounding factors affecting skeletal and UA dimensions and their impact on UA collapsibility [20, 21]. We also managed for the impact of BMI-related gender differences in the interpretation of inter-variable relationships (Table 3) and confirmed no inter-class differences in the male fraction (Tables 4 and 5). The present study realized that our CSP samples corresponded to non-obese young adult patients with severe OSA showing no gender differences from the non-CSP, and then characterized the polysomnographic profiles of them. It should be noted that various confounding factors cannot be perfectly adjusted for in every statistical

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analysis, and thus each condition-based outcome interpretation should be listed as a routine clinical practice for OSA patients.

This study has some limitations. (1) A referral bias in the sample collection may exist due to the retrospective single-center study design, (2) the sample size was insufficient to be generalized for phenotyping of all OSA patients, (3) imbalance in male/female fraction was inevitable due to the inherent higher prevalence of OSA in men, (4) CBCT images taken while standing/awake as a reflection of the airway during supine/sleep, and (5) the nonanatomical risk factors might have affected OSA severity in some cases. Therefore, the CSP may not be clearly differentiated from the complex phenotype including skeletal deformity.

Nonetheless, we exclusively conducted a phenotype-based multiperspective analysis to establish a clinically available guideline for the CSP of OSA patients in orthodontic clinics. Further studies with big-data samples are anticipated for an in-depth intrastructural characterization to be integrated with nonanatomical variables of OSA presentations to yield a more reliable phenotype-based precision approach for the OSA population.

Conclusion

From a multidimensional perspective, the CSP of OSA patients showed more severe OSA with more vulnerable nasal and pharyngeal airway features than the non-CSP patients, despite being younger and less obese. Specifically in the Class II pattern, the CSP_{3D} showed greater OSA severity than the CSP_{2D} indicating a significant impact of TSD on multivariable polysomnographic profiles.

Author contributions

Jae-Yon Roh (Data curation [Equal], Methodology [Equal], Software [Equal]), Nurdana Darkhanbayeva (Data curation [Equal], Investigation [Equal], Methodology [Equal]), Hye Kyu Min (Conceptualization [Equal], Supervision [Equal], Validation [Equal]), Kyung-A Kim (Formal analysis [Equal], Investigation [Equal], Validation [Equal], Writing—review & editing [Equal]), and Su-Jung Kim (Conceptualization [Lead], Formal analysis [Lead], Project administration [Lead], Supervision [Lead], Validation [Lead], Writing—original draft [Lead], Writing—review & editing [Lead])

Conflict of interest

We have no conflict of interest to declare.

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None to declare.

Disclosure

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Data availability

The underlying data are available within this article.

Supplementary material

Supplementary material is available at *European Journal of Orthodontics* online.

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